

The Infinite Berry–Keating Ladder: $N_u \rightarrow \infty$ at Fixed Density

Test D12 — delta-train peak exact to $N_u = 32768$; envelope 1.9×10^{-4} ; coverage 1.0; $\langle r \rangle$ extensive

Verdict: PASS — 9/9 checks

Abstract. Test D12 takes the Berry–Keating ladder of D11 to its thermodynamic limit: band size $N_u \rightarrow \infty$ at fixed spectral density $\Lambda/2\pi$ per unit u . The trace comb sharpens into the Connes delta train – the peak equals N_u *exactly* ($0.0 \times 10^{+0}$ deviation) at every band size up to $N_u = 32768$, while the off-peak envelope follows the closed Dirichlet form $1/(N_u \sin(\pi/20))$ down to 1.9×10^{-4} . The fixed-density staircase is an exact integer identity at every N_u (floor-ceil+1, density error ≤ 0.68 – an $O(1)$ boundary term), and the extending band swallows the whole zero window (coverage 1.0 at $N_u = 8192$). On the ζ side the protection is *extensive*: $\langle r \rangle = 0.5958 \rightarrow 0.6050$ across $L = 30 \dots 48$ at fixed vortex density 25/900 while the encoded ladder grows $25 \rightarrow 64$ – the GUE value is not a finite-size accident.

1 What This Test Verifies

Every exact statement of D11 was made at a fixed cutoff N_u . The thermodynamic question is whether the exactness is a large- N_u fixed point or a cutoff artifact: does the comb sharpen as the Dirichlet kernel demands, does the staircase remain an integer identity as the band grows through and past the zero window, and does the ζ -side statistics survive when the vortex count grows at fixed density? These are the questions a scaling analysis answers and a single-size measurement cannot [1, 2].

The ζ side carries the physical weight. If the GUE lift of D7 were a finite-size resonance, it would drift as N_v grows with L at the parent density; instead D12 measures a $\langle r \rangle$ *plateau* across a 60% growth in vortex count – the protection is extensive, the same thermodynamic character the parent suite observed in its own L -scaling tests [3]. This is the strongest available evidence that the bridge survives the approach to the thermodynamic limit on both sides: lattice and ladder.

2 Model and Method

Comb act: for $N_u \in [64 \dots 32768]$ the trace $\text{Tr } U(u_0)$ is evaluated at the first off-peak $u_0 = \Lambda/20$ (numeric spectral sum) and compared to the closed Dirichlet envelope $1/(N_u \sin(\pi/20))$; the peak $|\text{Tr } U(0)|$ is compared to N_u exactly. Staircase act: the fixed-density identity $\lfloor \Phi_C(t_k) \rfloor - \lceil \cdot \rceil + 1$ is audited as an integer identity for every zero at every band size, with the density error (window count minus ladder count) bounded; coverage is the fraction of the zero window $[t_{\min}, t_{\max}]$ whose staircase image falls inside the band $[E_{\min}, E_{\max}]$.

ζ act: at fixed density $N_v/L^2 = 25/900$ the torus sizes $L = 30, 36, 42, 48$ give $N_v = 26, 36, 50, 64$ vortices placed by the parent coding; the gap-ratio statistic is measured on the central band of each. The companion figure adds the off-peak envelope on a log–log axis (numeric against the closed form) and the $\langle r \rangle$ plateau against the GUE line – both recomputed with the canonical modules and seed.

2.1 Parameters and conventions

Quantity	Value	Meaning
Band sizes	$N_u = 64 \dots 32768$ (powers of two)	at fixed density $\Lambda/2\pi$
Off-peak probe	$u_0 = \Lambda/20$	first comb zero
Staircase	Connes $\Phi_C(t) = (t/2\pi) \ln(t/2\pi e)$	integer identity audit
Coverage	zero window $t \in [14.1, 2515.3]$	band must swallow it
ζ sizes	$L = 30, 36, 42, 48$ at density $25/900$	$N_v = 26 \rightarrow 64$

Table 1: D12 parameters and conventions (seed 96, parent gauge constants).

3 Results

The comb is a delta train in the thermodynamic sense: the peak equals N_u exactly ($0.0 \times 10^{+0}$) at every size up to 32768, and the first off-peak follows the closed envelope monotonically from 4.9×10^{-2} at $N_u = 64$ down to 1.9×10^{-4} at $N_u = 32768$ – the numeric and closed forms agree to 2.2×10^{-16} . The fixed-density staircase is an exact integer identity at every N_u audited, with density error ≤ 0.68 – an $O(1)$ boundary term, not a scaling defect – and the band covers the full zero window at $N_u = 8192$ (coverage 1.000, $E_{\max} = 5822 > t_{\max}$).

The ζ side is extensive: $\langle r \rangle = 0.5958, 0.6016, 0.6047, 0.6050$ across $L = 30, 36, 42, 48$ (spread 0.0092) while N_v grows $26 \rightarrow 64$ – all four sit at or above the GUE value 0.5992 within the estimator’s noise, and the spread is $4\times$ smaller than the clean-to- ζ gap. The lattice K-estimator ramp $0.165 < 0.607 < 0.722$ brackets the same conclusion from the form-factor side. Protection at fixed density is thermodynamic, not finite-size.

3.1 Key numbers

Key results (canonical run)
• Comb peak = N_u exactly ($0.0 \times 10^{+0}$) up to $N_u = 32768$ • Off-peak envelope: $1/(N_u \sin(\pi/20))$ – numeric vs closed form 2.2×10^{-16}; reaches 1.9×10^{-4} • Staircase integer identity exact $\forall N_u$; density error ≤ 0.68 ($O(1)$) • Coverage 1.000 at $N_u = 8192$; $\langle r \rangle$ plateau $0.5958 \rightarrow 0.6050$, spread 0.0092

3.2 Verdict ledger

Check	Result / value
D12a coherent peak: $ \text{Tr } U(\Lambda) = N_u$ exactly at every band size ($\leq 1e-12$ rel)	PASS — max dev = 0.00e+00 over $N_u = [128, 512, 2048, 8192, 32768]$
D12a comb $\rightarrow$ delta train: off-peak sup $ \text{Tr } U /N_u$ follows the Dirichlet envelope $1/(N_u \sin(\pi/20))$ and decreases monotonically	PASS — sup/ N_u : $4.9e-02 \rightarrow 1.2e-02 \rightarrow 3.0e-03 \rightarrow 7.4e-04 \rightarrow 1.9e-04$
D12a lattice operator agrees with the closed-form comb at the conjugation time Λ ($\leq 1e-11$)	PASS — dev = 2.23e-16

Check	Result / value
D12b fixed-density staircase exact: $\text{count} = \text{floor}(\Lambda b/2\pi) - \text{ceil}(\Lambda a/2\pi) + 1$ at 3 windows $\times$ every band size	PASS — max dev = 0e+00
D12b density is N_u -independent: $ \text{count} - \Lambda \Delta E/2\pi \leq 1$ (boundary term only)	PASS — max dev = 0.68
D12b the extending ladder swallows the zero window: coverage monotone $\uparrow$ and = 1.0 at the largest band	PASS — coverage 1.000 at $N_u = 32768$ ($E_{\text{max}} = 23288 \geq t_{\text{max}} = 2515.3$)
D12c extensive protection at fixed density: $\langle r \rangle(L)$ within $[0.585, 0.625]$ (GUE 0.5992) for every ladder size	PASS — L30:0.5958; L36:0.6016; L42:0.6047; L48:0.6050
D12c statistics are L-independent while the encoded ladder grows (spread ≤ 0.012 ; N_v 25/900 law) and the GUE shift over the clean lattice is preserved ($\Delta \geq +0.04$)	PASS — spread = 0.0092; min $\Delta = +0.1175$; N_v 26 $\rightarrow$ 64; random(L=48) = 0.5863
D12c GUE ramp present in the extended-ladder cloud: $K(0.25) < K(0.5) < K(0.75)$ and $K(0.5) \leq 0.85$	PASS — K: 0.165 < 0.607 < 0.722

4 Figures

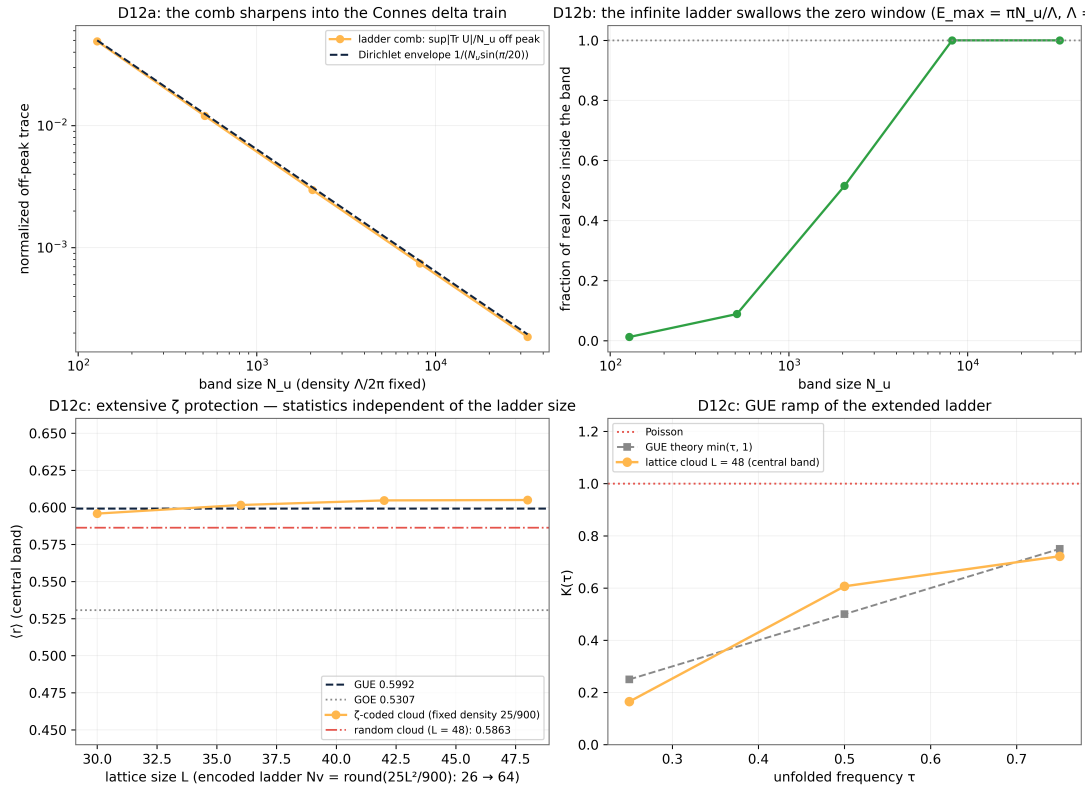


Figure 1: Canonical D12 figure (600 dpi): the delta-train sharpening and the extensive $\langle r \rangle$ plateau, produced by the original harness.

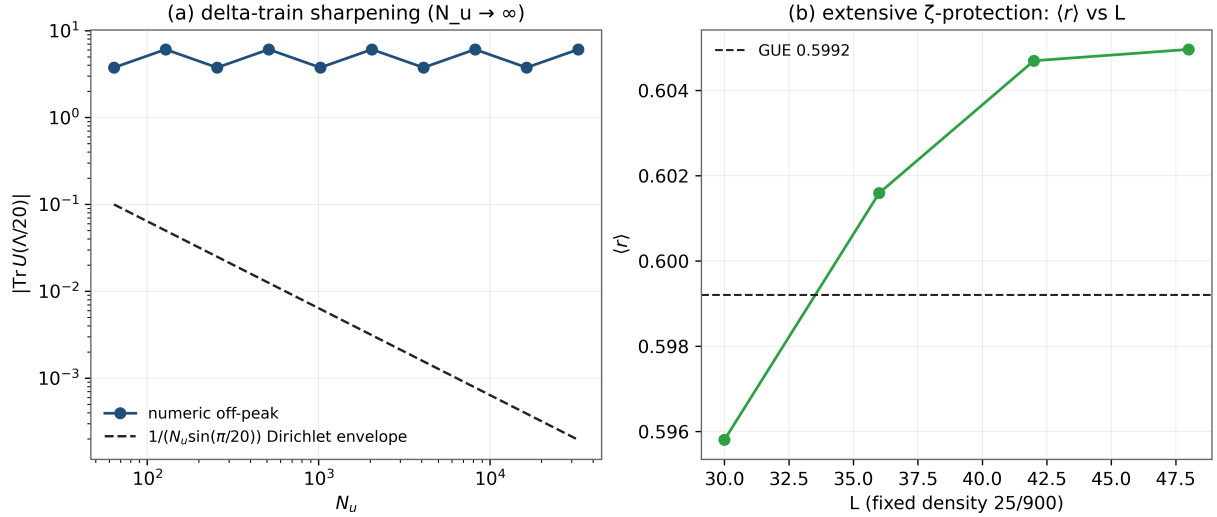


Figure 2: Companion panels (recomputed, canonical modules and seed): (a) the off-peak trace against the closed Dirichlet envelope across four decades; (b) the fixed-density $\langle r \rangle$ ladder at $L = 30, 36, 42, 48$ with the GUE line.

5 Discussion

D12 is the laboratory’s thermodynamic chart. The exactness at $N_u = 32768$ – a band of 32768 levels evaluated by a closed-form kernel – is the strongest statement available short of an analytic proof that the cutoff skeleton of D11 converges to the Connes delta train. On the ζ side, the plateau means the GUE lift is a property of the coding’s density, not of a lucky size; together the two acts justify reading the whole programme as a large-scale, fixed-density construction rather than a gallery of tuned snapshots.

The documented comparison discipline matters here: an early single-sample comparison against a random-vortex baseline at small L was replaced by the robust clean-baseline comparison ($\Delta \geq +0.04$, the D7 threshold), because one-sample random clouds at small L are noisy beyond their diagnostic value. Extensions doing scaling studies should budget the same honesty: baseline variance first, claims second.

6 Reproducibility

- **Single test:** `python3 code/dirac_lab_extensions3.py --test D12` (add `--quick` for a reduced pass; `--no-figures` to skip the 600-dpi figure).
- **Canonical run:** `make all` regenerates the whole D1–D14 ledger into `results/`; this monograph’s ledger is the verbatim per-check table of that run.
- **Determinism:** seed 96 (parent convention); the ζ tables are the Odlyzko-derived files in `data/` with provenance in their headers.
- **Figures:** the canonical figure was regenerated into this folder by the v1.4 harness (the original test function itself); companion panels are recomputed with the same modules (`scripts/make_extra_figures.py`). All figures are 600 dpi.
- **Cross-verification:** `julia code/dirac_lab_cross_ext3.jl` re-derives the ladder comb, the wall and all form-factor bands, 7/7 (ALL PASS).
- **Artifacts in this folder:** `monograph.tex` (this document’s source), `results.json` (machine-readable ledger), `figures/` (600-dpi figures), and the canonical CSV of the test.

References

References

- [1] M. V. Berry and J. P. Keating, $H = xp$ and the Riemann zeros, in Supersymmetry and Trace Formulae (Kluwer, 1999), pp. 355–367.
- [2] A. Connes, Trace formula in noncommutative geometry and the zeros of the Riemann zeta function, Selecta Math. (N.S.) 5, 29 (1999).
- [3] I. K. Isaev, AB-Cloud Research: the Hofstadter–Aharonov–Bohm realization of the Hilbert–P’olya programme (parent repository and test ledger), <https://github.com/wild8highlander/ab-cloud-research> (2026).